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BEYOND DETECTION: AN INFORMED DEPLOYABLE SOLAR FLARE FORECASTING AND REAL TIME X-RAY FLUX PREDICTION SYSTEM

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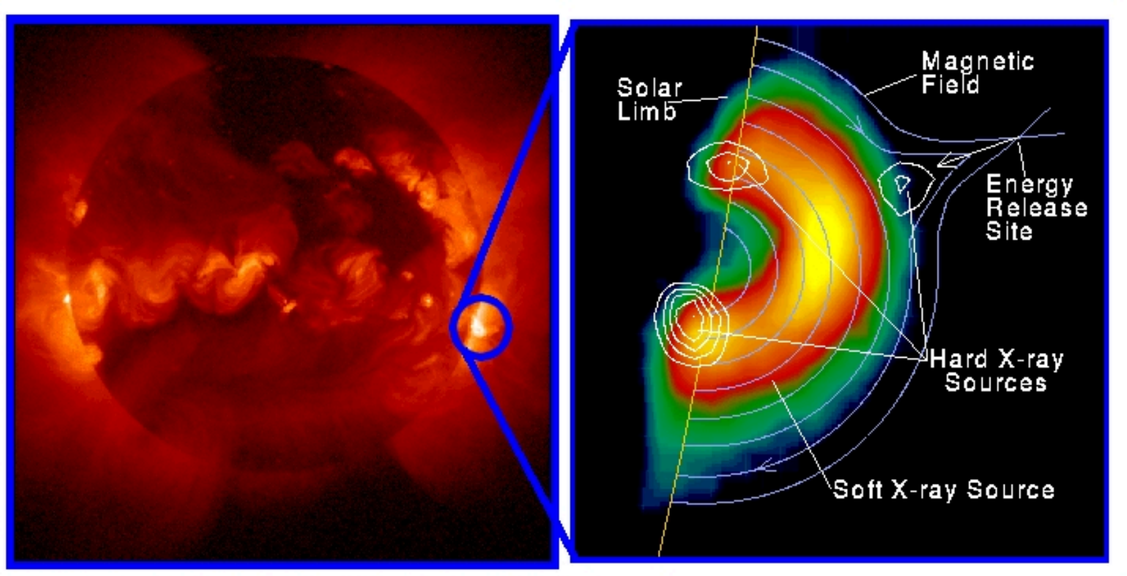
INTRODUCTION

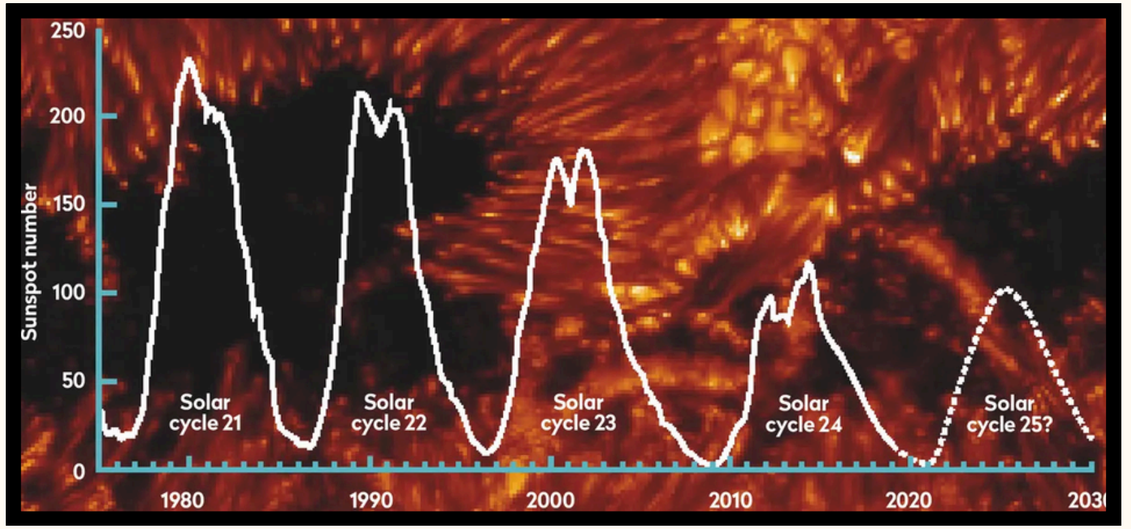
Solar Flares:

- Explosions of energy and electromagnetic radiation that arise from the **entanglement and bursting of magnetic loops** on the sun's coronal layer

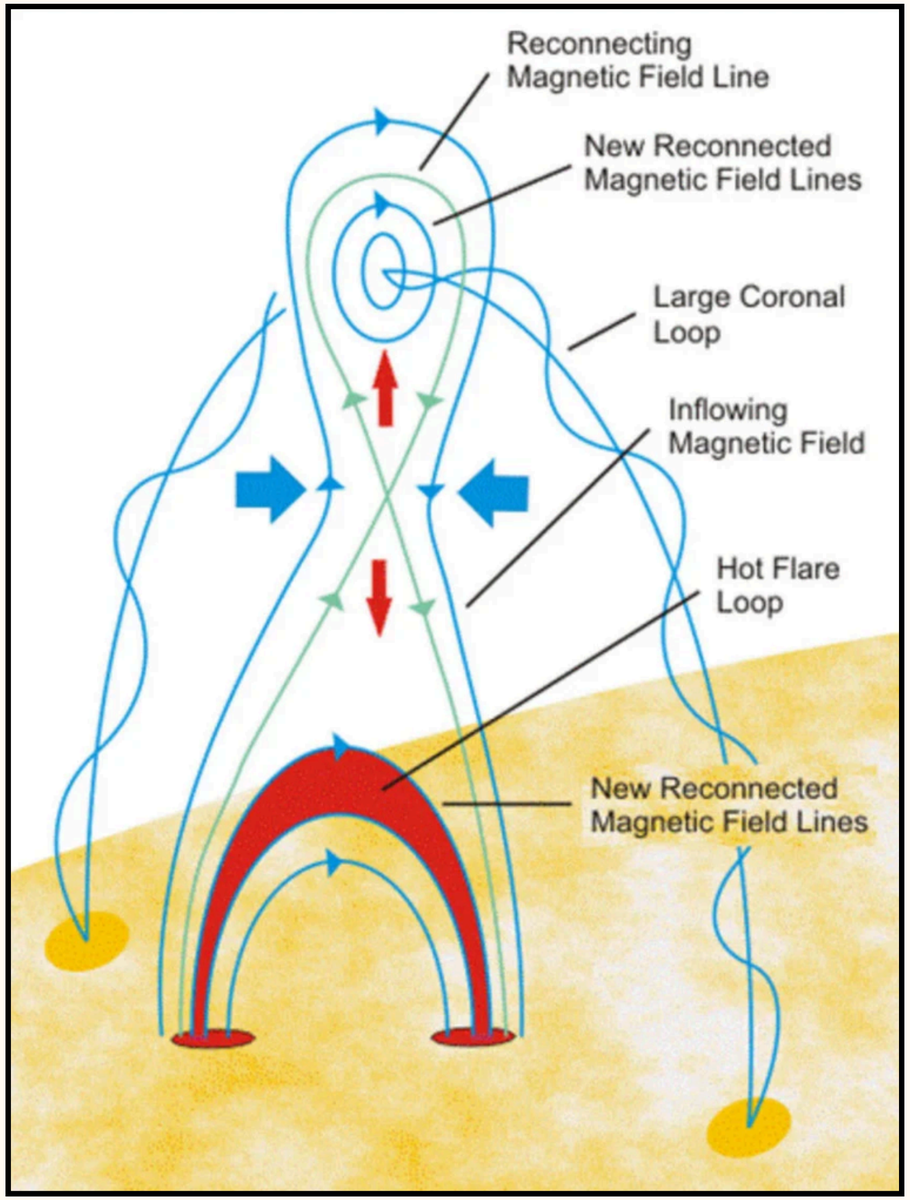
- Classified by the peak strength of energy released
  - A-Class, B-Class, C-Class, M-Class, X-Class
  - Solar activity above C-1.0 in magnitude can cause **significant geomagnetic disturbance**

- **2025 as the solar maximum** of the 11-year Schwabe Cycle describing solar activity





INTRODUCTION



Magnetic Reconnection:

- Rapid rearrangement of magnetic field lines in highly stressed solar active regions

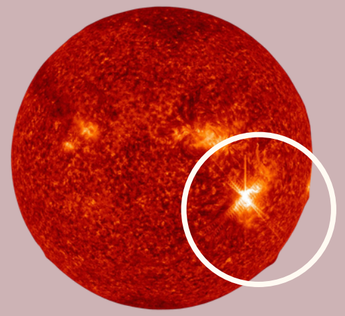
- **Energy released as:**
  - X-ray and extreme ultraviolet radiation
  - High-energy particle acceleration
  - Plasma and shock waves

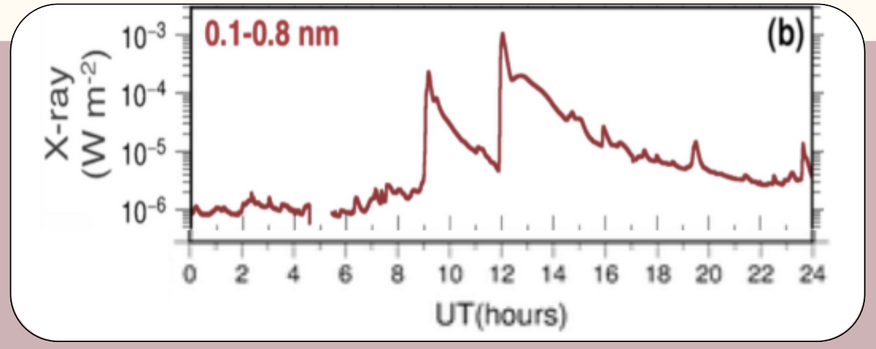
- Electromagnetic radiation can reach us in a matter of **8 minutes** → **MOST IMMEDIATE THREAT**

INTRODUCTION

Risks of Solar Flares:

- X-ray and gamma radiation emitted by strong flares can **ionize the ionosphere** and:
  - disrupt or destroy satellite electronics, communication systems, and power grids
  - pose serious risks to nearby astronauts or high-altitude flight crews
- Solar flares accounted for around **20% of all satellite failures** from 1996-2010
- Estimated that the global economy loses around **$17 billion** during a single solar storm





X-ray irradiance detected by GOES-13 on 9-6-2017 during a X9.3 flare.

Current Flare Forecasting Techniques:

Poisson Flaring Probabilities:

- Statistical method based on historical flare catalogs and trends
- **Calculates likelihood** of observing a certain number of solar flares within a given time period

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Where:
$x =$ number of successes
$\lambda =$ rate of success
$e =$ Euler's number (≈ 2.71828))

Limitations

Poisson flaring probabilities are **limited by their assumptions** of independent flare rates, lack of physical knowledge, and moderate accuracy with high false alarm rates.

INTRODUCTION

Current Flare Forecasting Techniques:

Machine Learning Techniques:

- **Predicts** whether a solar flare will occur within a given time period (e.g. 6 hours, 12 hours, 24 hours)
- Can rely on solar image datasets or physical feature datasets
- Ex. Neural Networks, VGG-16, AlexNet, ResNet, SqueezeNet

Limitations

1

Excessive data volume and computational cost limit scalability and operational efficiency

- Long processing times **restrict accessibility** and **practical deployment**

2

Focus on classification without forecasting continuous radiation energy emissions:

- **Soft X-ray Flux:** indicator of intensity of low-energy X-ray radiation emitted by the Sun
  - Map flare impact and strength evolution

ENGINEERING OBJECTIVE

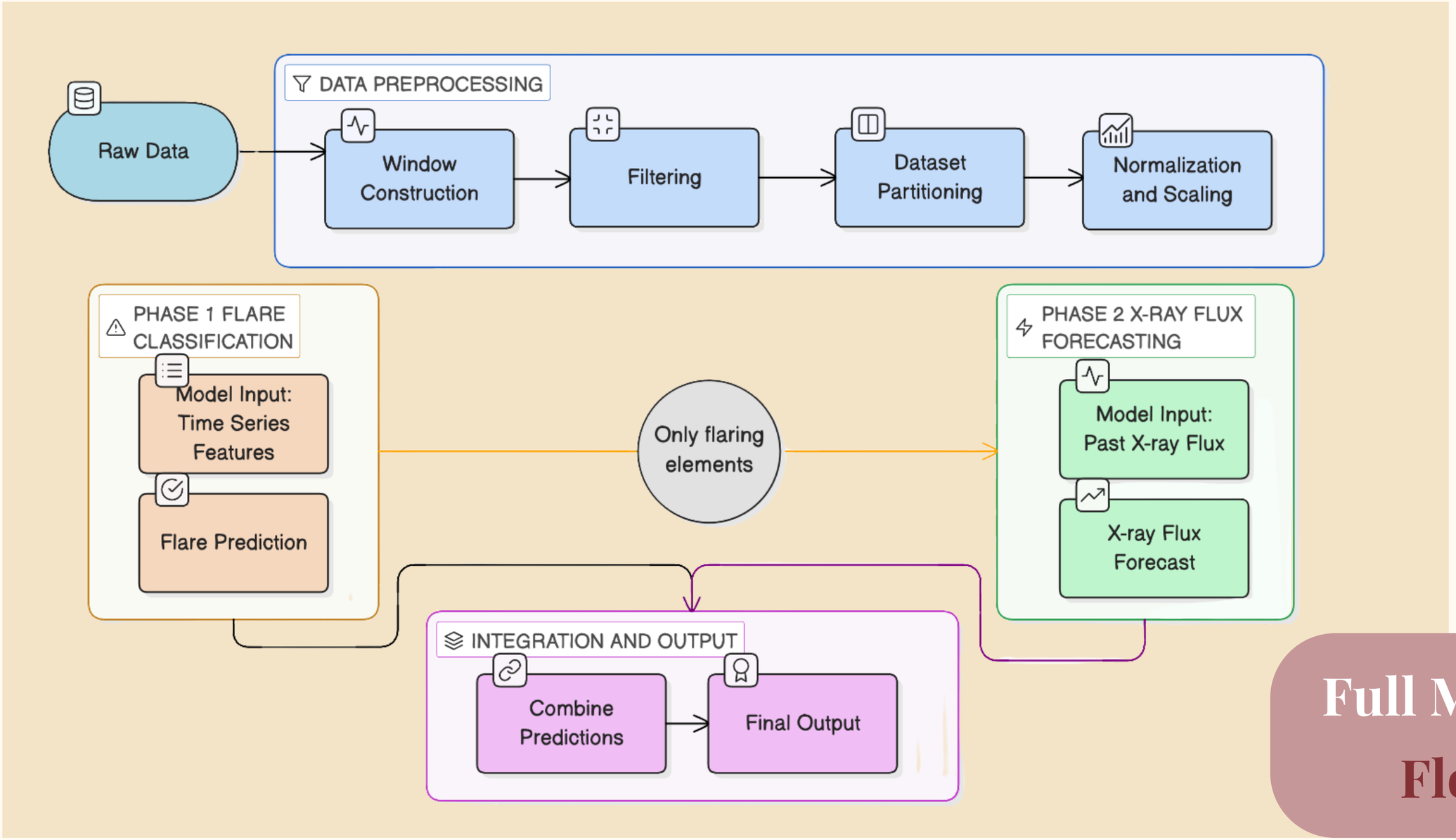
Develop an integrated flare forecasting framework capable of:

1. Forecasting **continuous** X-ray flux evolution on top of flare occurance classification
2. Generating **comprehensive** physics-informed reports based on magnetic features and flux curves
3. Maintaining **computational** and data efficiency while balancing predictive performance
    - Able to run on standard and cheap GPU computing systems

RESEARCH OBJECTIVE

1. Evaluate the impact of data limitations on the modeling and interpretation of solar flare dynamics
2. Investigate trade-offs and identify optimal hyperparameter configurations that maximize accuracy, efficiency, and data usage
3. Examine approaches to support reliable real-time forecasting under practical constraints

METHODOLOGY



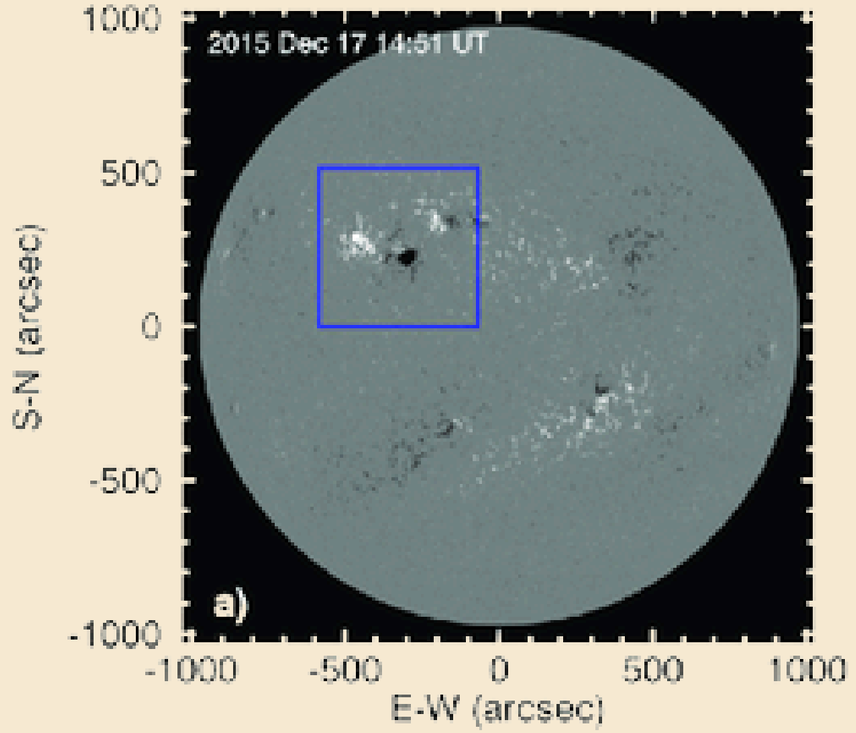
Full Methodology Flow Chart

METHODOLOGY

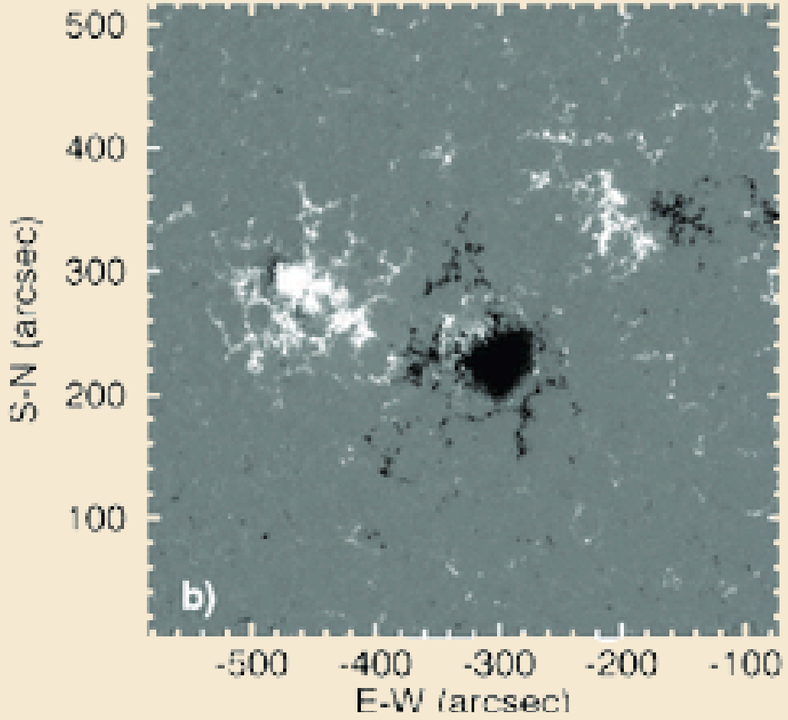
Magnetogram Images:

- Maps the **strength and polarity of sun's photospheric magnetic field** within an identified solar Active Region (AR)
- Recorded by the Helioseismic and Magnetic Imager aboard NASA's Solar Dynamics Observatory
- Used to extract quantitative physical features: magnetic flux, polarity inversion lines, current helicity, etc.

full disk magnetogram



active region patch magnetogram

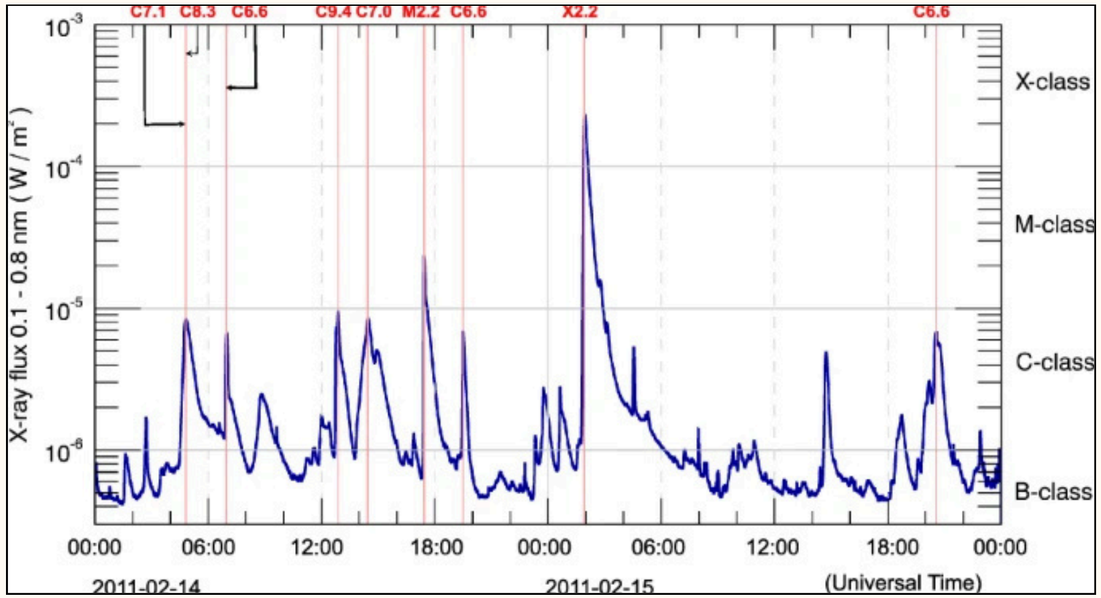


METHODOLOGY

Data Acquisition:

- **GOES Soft X-ray flux (1 to 8 Angstroms)**
- **Spaceweather HMI Active Region Patch Physical Descriptors**

- Each time step represented a 12 minute average
- Data spanned from May 2010 to December 2018



GOES X-ray Flux

Cleaning and Filtering:

- Quality flag **filtering** of invalid observations
  - X-ray Flux Quality Index

- **Removal** of data elements with NaNs and physically-invalid values

- Only solar regions within **70° from solar central meridian** were included
  - Prevents projection distortion

- **Log transformation and scaling**

$$F_{\log} = \log(F + \epsilon)$$

  - Computed on only the training set to prevent information leakage
  - Applied to testing and validation set

METHODOLOGY

Input and Output Time Series Construction:

- Constructed fixed-length feature windows of 3 hours pre flare and 1 hour post flare
- **Two-pronged sampling strategy:**
  - **Targetted sampling:** centered on flare onsets
  - **Random sampling:** general flare associated intervals
- Assigned class labels: 0 - no flare, 1 - m flare, 2 - x flare

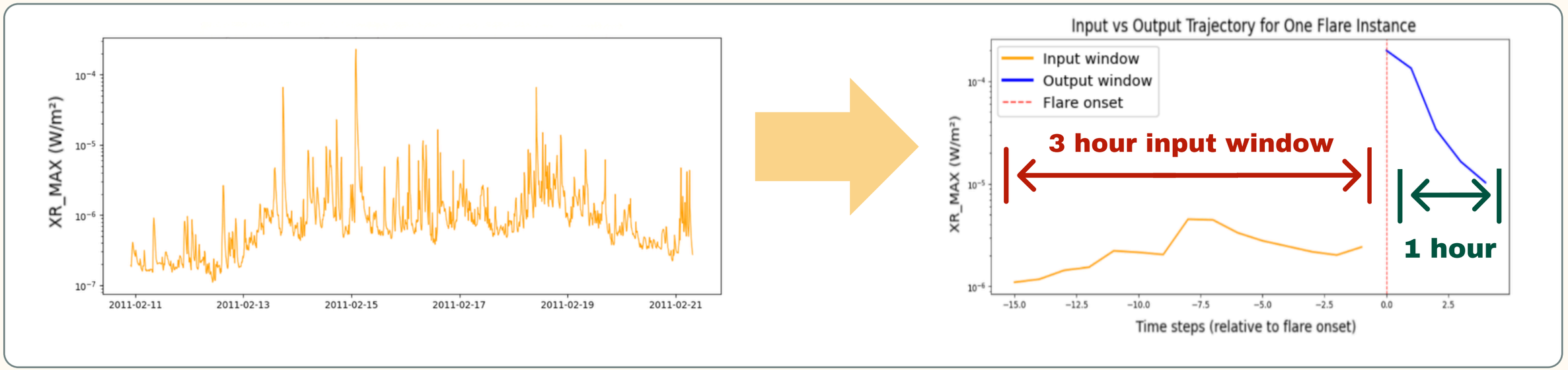
DATASET SUMMARY

Total samples: 7,976

Non flare class: 62%

M-flare class: 36%

X-flare class: 2.26%



METHODOLOGY

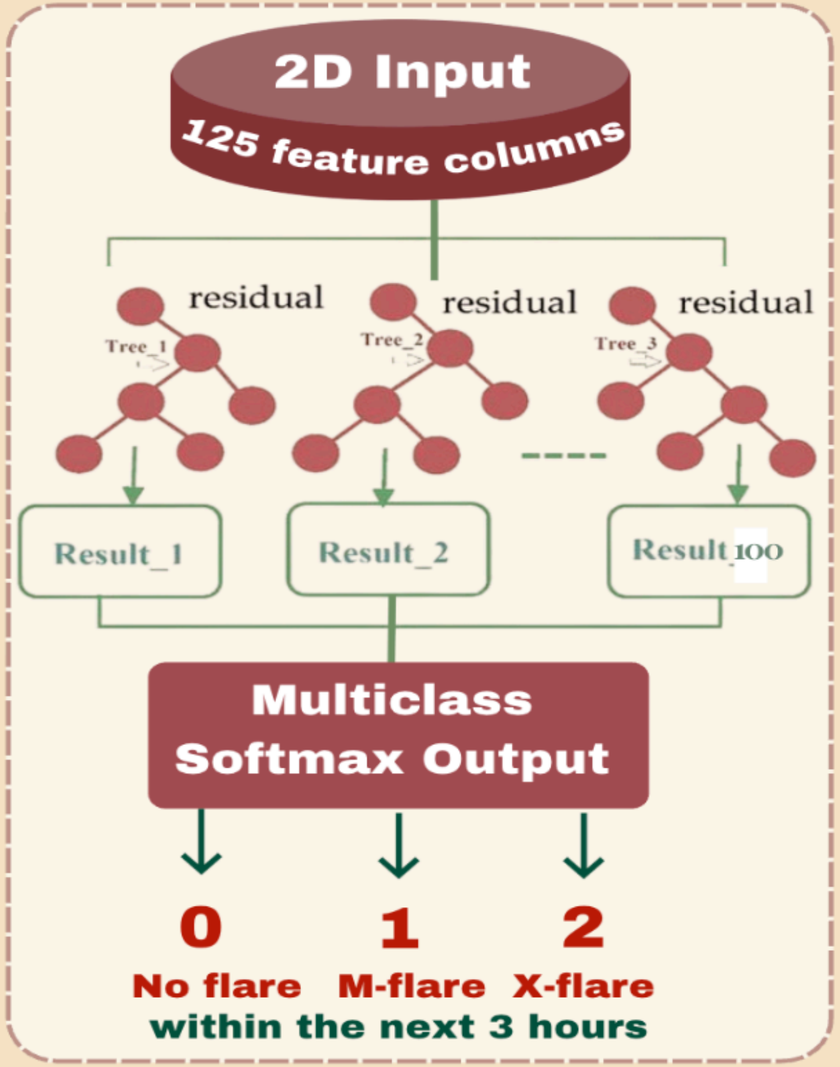
Phase 1: Classification

INPUT: 125 physical features describing the solar region
OUTPUT: Predicted class of flare within the next three hours

XGBoost Model:

- **Tree-based approach**
  - Sequentially adds decision trees and learns through **Gradient Boosting**

- Model hyperparameters:
  - n_estimators = 100
  - learning rate = 0.1
  - max_depth = 6
- Optimized by a **multinomial loss function**

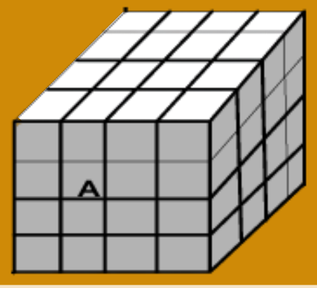


METHODOLOGY

Phase 1: Classification

Temporal Feature Aggregation:

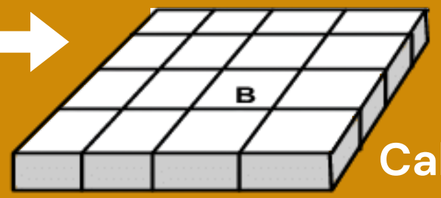
- Flattened time series into a 2D dataset of engineered statistical features for XGBoost Input



(Samples, Time, Features)

Per-feature temporal statistics

- mean, std, max, min, last value



(Samples, Calculated Features)

Feature Importance:

- Quantified total metric gain of each physical feature
- Calculated **Shapley Additive Explanations (SHAP)**
  - Contribution of each feature to final model prediction

$$Gain = \frac{1}{2}\left(\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right) - \gamma$$

METHODOLOGY

Phase 2: Regression

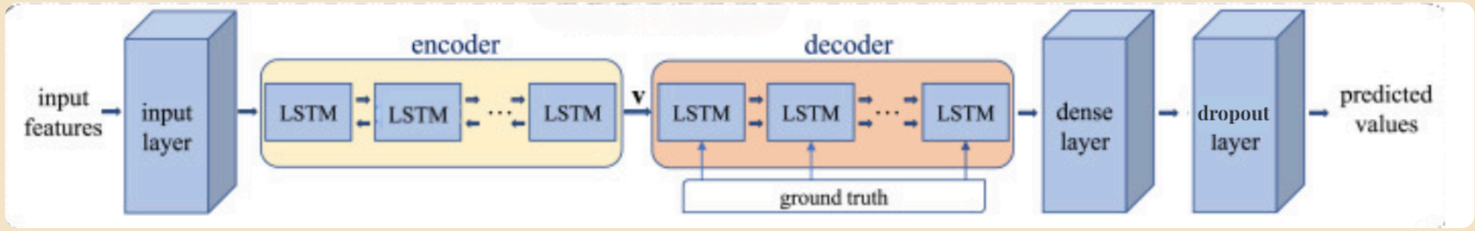
INPUT: 3 hour time series of pre-flare X-ray Flux
OUTPUT: 1 hour time series of post-flare predicted X-ray Flux

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Model Framework:

- Seq2Seq LSTM
- Encoder and Decoder
- Attention-based Concatenation

Seq2Seq Long Short Term Memory Algorithm



Algorithm Specifics:

- Optimized based on Huber Loss Function
- Early Stopping with a learning rate reduction
- Autoregressive inference model for predictions

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta\left(|y - \hat{y}| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases}$$

METHODOLOGY

Phase 2: Regression

Training-Specific Implementations

Teacher Forcing

- Iteratively fed ground-truth target sequence to continue with during training

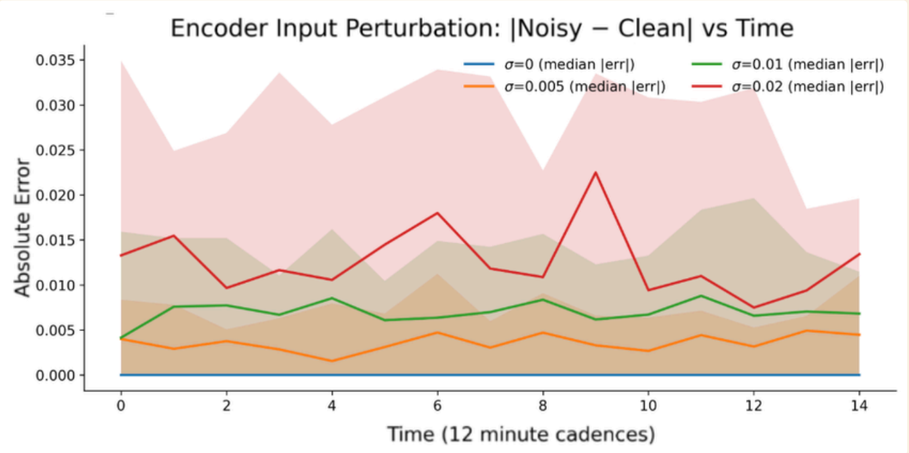
Grid Search

- Optimized hyperparameters through RMSE
- 5-fold cross-validation

Gaussian Noise Augmentation

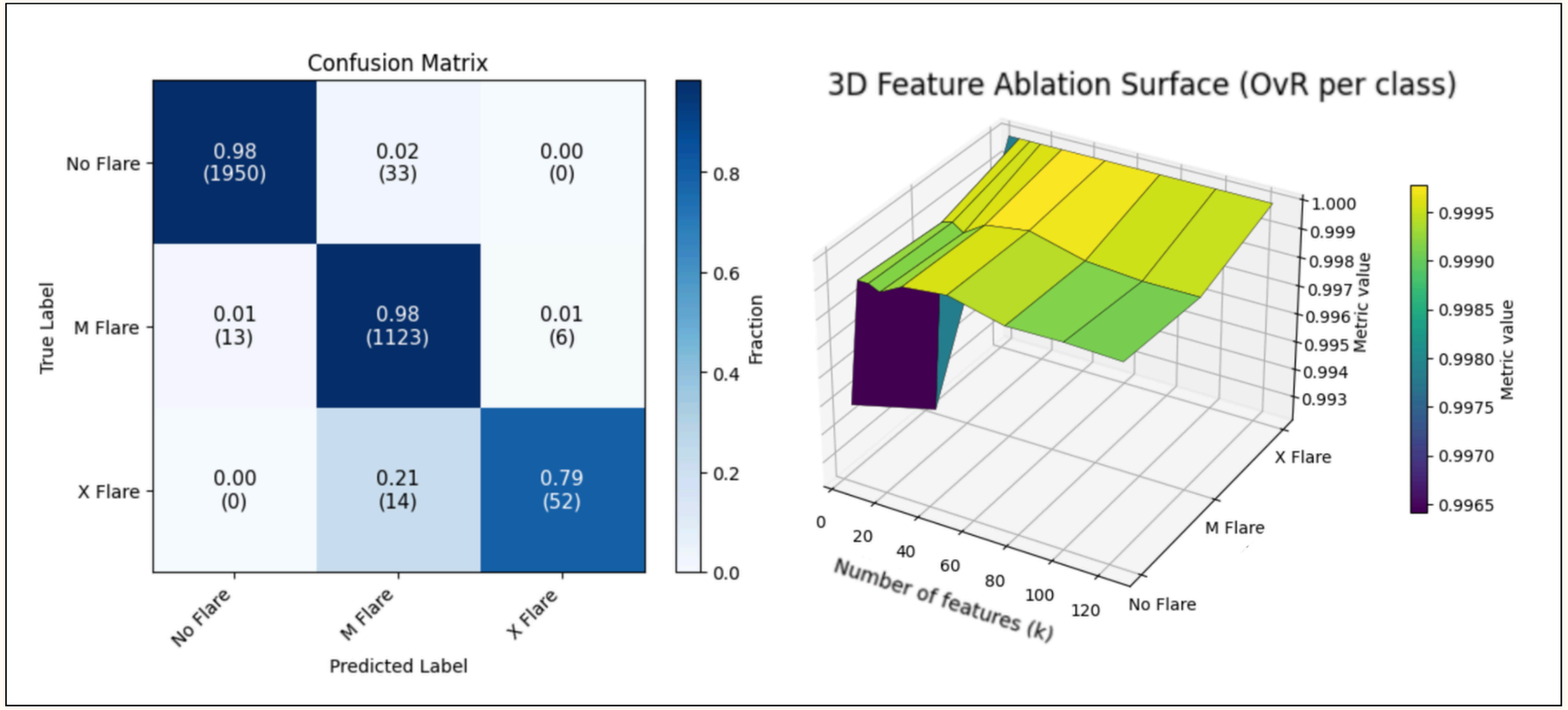
- Addresses issue of sparse flare data
- Improves robustness to uncertainty

$$x' = x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$
$$\sigma = 0.02$$



RESULTS

Phase 1 Classification Testing Set Performance:



Metrics

Overall accuracy:

- 98%

No flare-class:

- **98%**

M-class

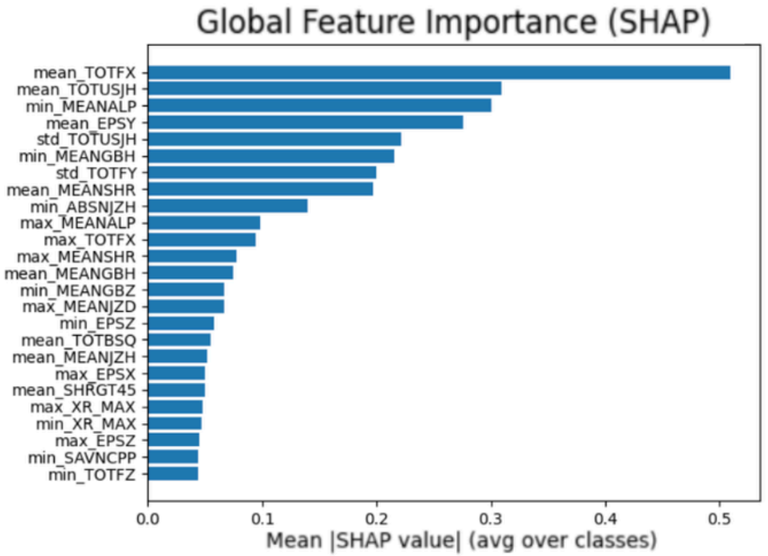
- **98%**

X-class

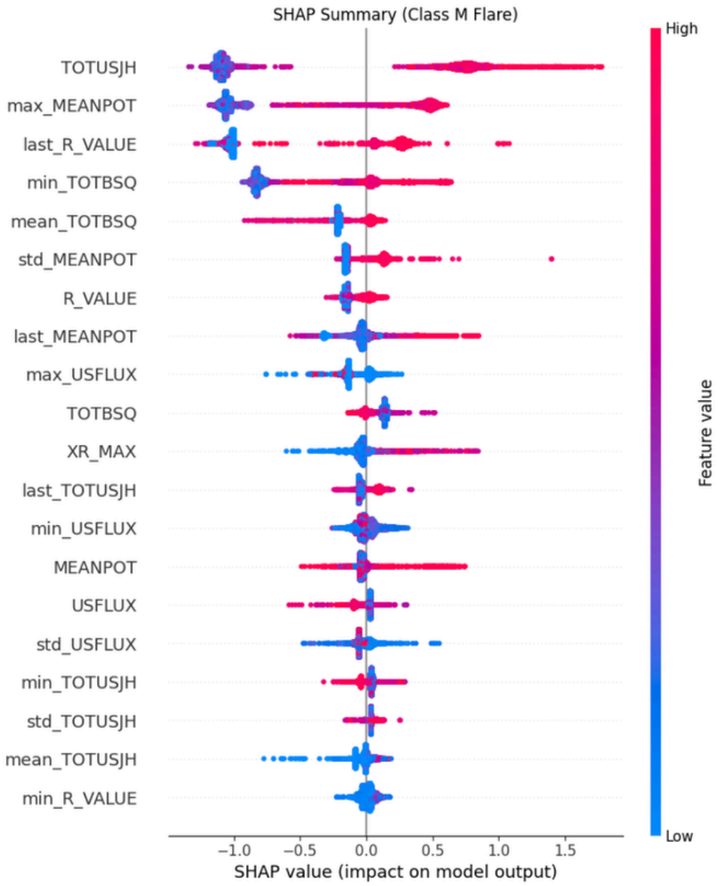
- **79%**

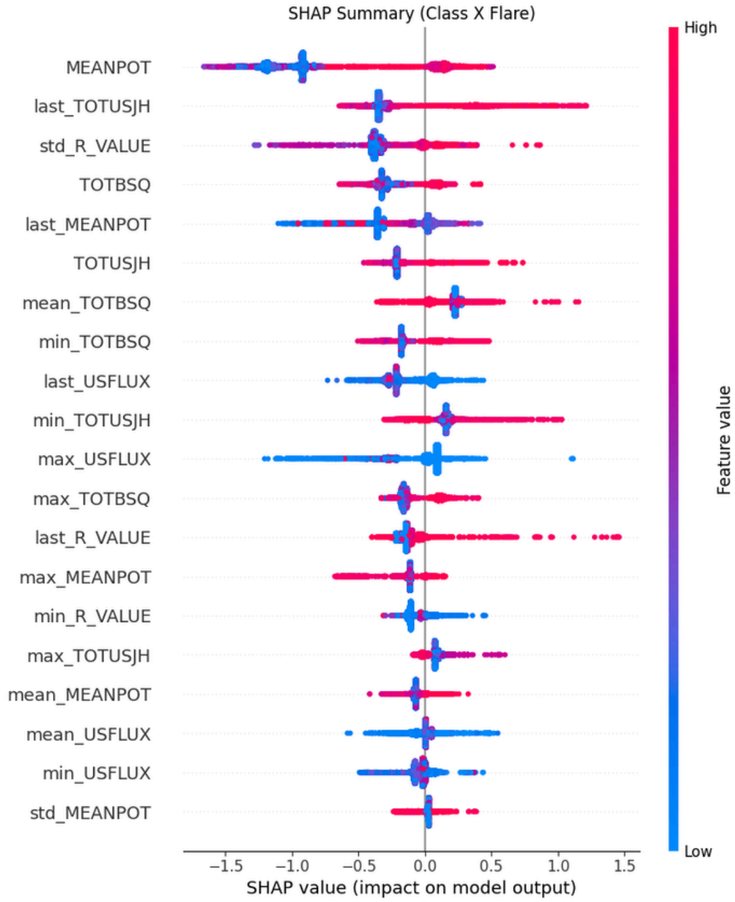
RESULTS

Feature Importance Analysis:



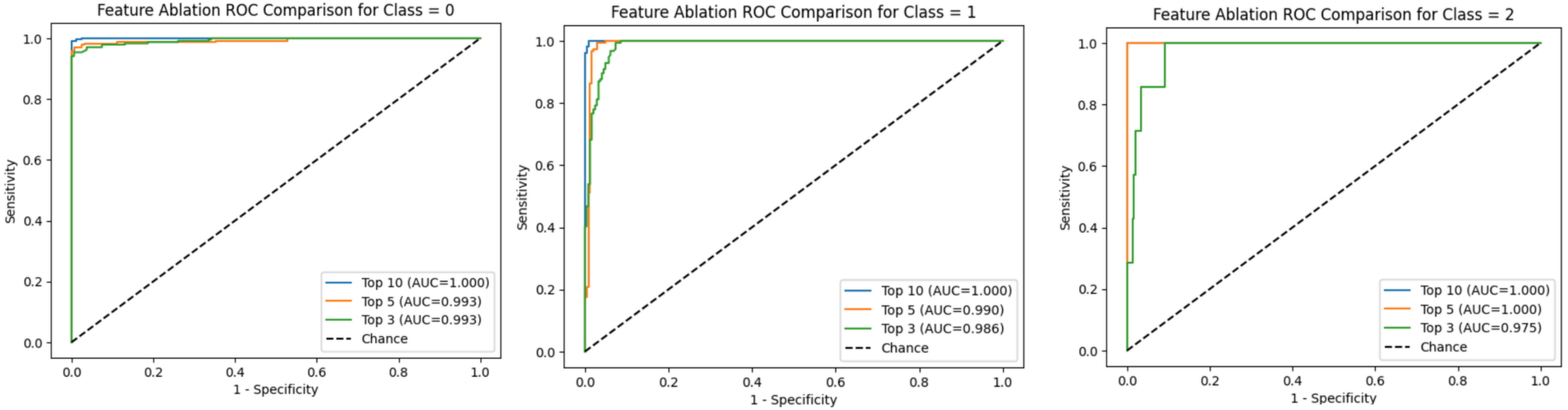
Global Feature Importance ranked on Shapley Additive Explanations values.





RESULTS

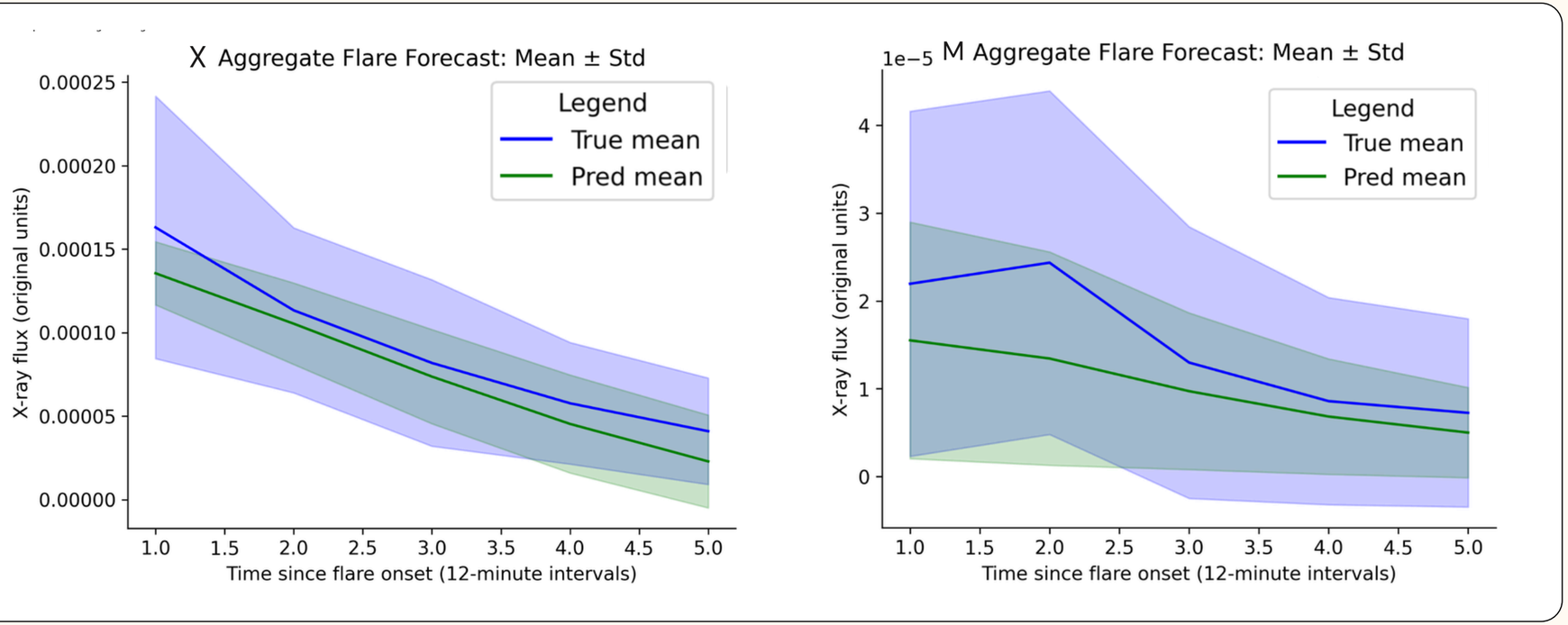
Feature Ablation ROC Comparisons:



Even with a **limited subset of features**, the **classifier retains high sensitivity and specificity** across all classes, highlighting both the robustness of the model and the effectiveness of the feature selection process.

RESULTS

Phase 2 Regression Testing Set Performance:



Metrics

X-class: RMSE = 5.56×10^{-5} (~16% of peak flux)

- $\log_{10}$ RMSE = 0.44

M-class: RMSE = 1.40×10^{-5} (~16% of peak flux)

- $\log_{10}$ RMSE = 0.38

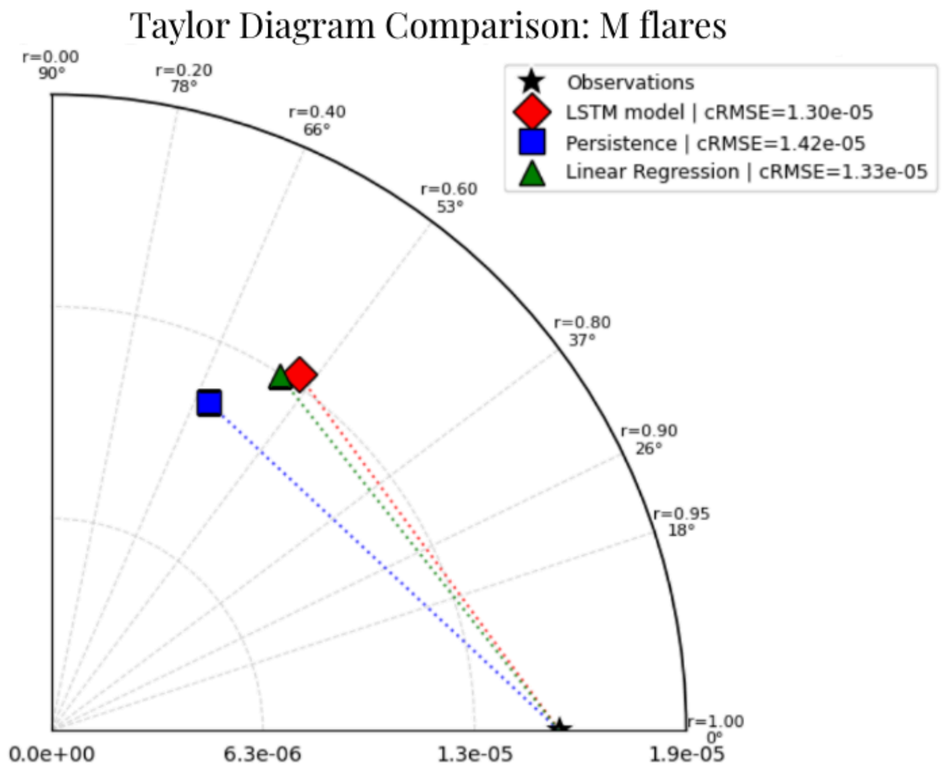
Efficiency: Early stopping after 73 (X) / 68 (M) epochs with stable validation loss

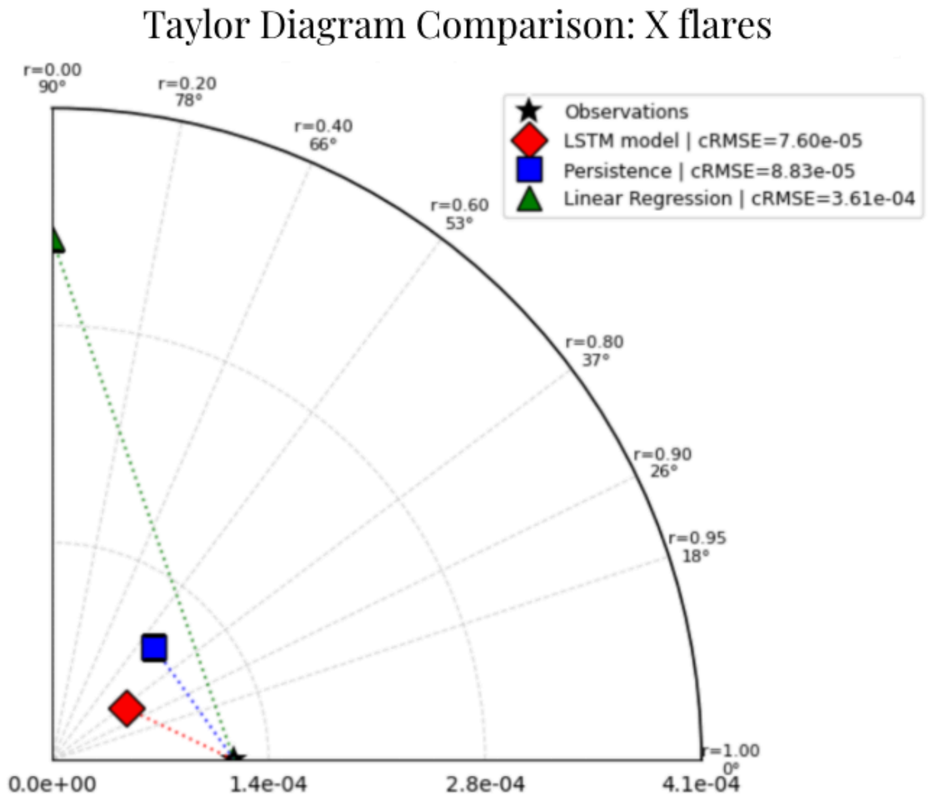
Inference: 0.49 s (X) / 0.59 s (M) per unseen input

RESULTS

Phase 2 Regression Testing Set Performance:

Taylor Diagrams with Comparisons to Persistence and Linear Regression Baselines





IMPACT AND APPLICATIONS

1 

Processes around **120 flare forecasts** per minute, enabling rapid real-time monitoring of solar activity

2 

Identifies **high-risk active regions early** and focuses detailed forecasting only where flares are likely

3 

Provides alerts **3 hours** in advance and continuously forecasts the next **hour** of X-ray flux

4 

Supports applications in **space-weather forecasting** that can help protect satellites, communication systems, and power infrastructure

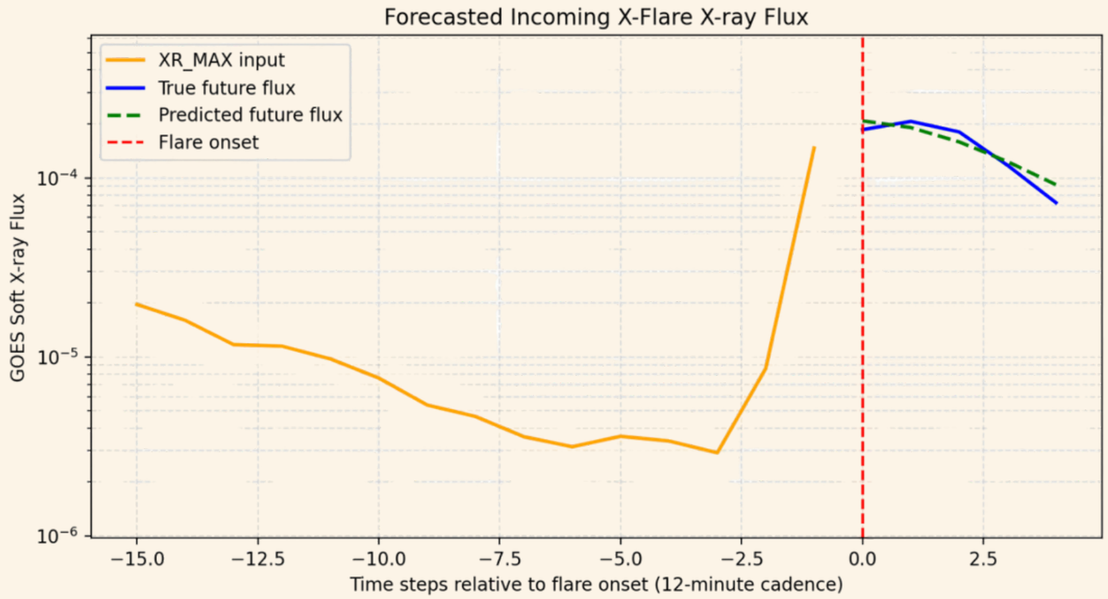
IMPACT AND APPLICATIONS

Phase 1: Flare Classification

Predicted Class	True Class	Probability (of predicted class)
X Flare	X Flare	0.952

Phase 2: X-Flare Flux Forecasting

Forecast Horizon	RMSE	Inference Time
60 minutes	1.799624e-05	156.2 ms



Real-Time Demo Created on Streamlit:

FURTHER RESEARCH

1. Extending the data to include magnetogram images and using **hybrid CNN-LSTM** architectures.

2. Extend the framework to include **CME analysis and reporting**, enabling joint study of flares and associated eruptions.

3. Exploring **Physics Informed Neural Networks (PINNs)** to set magnetohydrodynamics constraints.

4. Adding **more precautions to window-creation** processes to ensure validity

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THANK YOU

Please email **niu.isabelle@gmail.com**
for any remaining questions!

